

ADR-001: User Authentication and Authorization System

Status: Proposed

Date: October 3, 2025

Context

We need to implement a robust authentication system that allows internal users to securely log in and access content based on their roles, such as admin, editor, or viewer. This system will be built using Supabase for authentication and user management, ensuring a seamless registration and login experience, while Contentstack will manage the content that users can create, edit, or view. By establishing clear role definitions and permissions, we aim to enhance collaboration, streamline workflows, and ensure compliance with data protection regulations.

Key Non-Functional Requirements (NFRs)

- **Performance:** Ensure that Supabase handles database queries efficiently, with response times for CRUD operations generally under **200 ms** during peak usage, to maintain a responsive user experience.
- **Scalability:** Support an increasing number of concurrent users and data volume without significant performance degradation. Supabase should allow for horizontal scaling to accommodate growth in user registrations and content interactions.
- **Availability:** Target high availability with an uptime of **≥ 99.5%**. Implement failover strategies and redundancy within Supabase services to ensure continuous access to authentication and database functionalities.
- **Security:** Enforce strong security measures, including secure authentication protocols (e.g., JWT tokens), role-based access control (RBAC) for data access, and encryption of sensitive data at rest and in transit to protect user information.
- **Maintainability:** Ensure that the Supabase configuration and database schema are well-documented and modular, allowing for easy updates and modifications. Use version control for database migrations to facilitate smooth transitions during feature enhancements.
- **Observability:** Implement monitoring and logging for Supabase services to track performance metrics, user authentication events, and error rates. Provide dashboards for real-time insights and alerts for any anomalies or performance issues, enabling quick response to potential problems.

Options

1. Build Custom Authentication System with Supabase (Chosen)

- a. **Pros:** Centralized user management, robust security features, customizable role-based access control, and seamless integration with Contentstack for content management.
- b. **Cons:** Requires significant engineering effort, ongoing maintenance, and potential complexity in implementation.

2. Adopt an Off-the-Shelf Authentication Solution

- a. **Pros:** Faster initial deployment, reduced engineering overhead, and established security protocols.
- b. **Cons:** Limited customization options, potential integration challenges with Contentstack, and may not fully meet specific organizational needs.

3. Utilize a Hybrid Approach with Supabase and Third-Party Tools

- a. **Pros:** Leverage the strengths of Supabase for core authentication while using third-party tools for specific functionalities (e.g., analytics, user management).
- b. **Cons:** Increased complexity in managing multiple systems, potential integration issues, and reliance on external services for critical features.

Decision

We will pursue Option 2: Build a custom authentication system using Supabase. This approach ensures that we have a centralized and secure user management solution that aligns with our content management practices in Contentstack. It provides the flexibility to implement role-based access control tailored to our specific needs while adhering to strict non-functional requirements around performance, scalability, security, and maintainability. By leveraging Supabase, we can create a robust authentication experience that enhances user engagement and streamlines collaboration across our teams.

Consequences

Positive:

- **Centralized Authentication:** A unified system for user management, improving efficiency for all users accessing the CMS.

- **Enhanced Security:** Stronger security measures through role-based access control, reducing the risk of unauthorized access.
- **Improved User Experience:** A streamlined and intuitive login process that encourages user engagement and adoption.

Negative:

- **Initial Development Time:** The initial setup and development phase may lead to delayed timelines for overall project delivery.
- **Potential Scalability Challenges:** As the user base grows, unforeseen scalability issues may arise, necessitating further modifications or optimizations.
- **Limited External Support:** Custom solutions may face challenges in receiving timely support from the community or vendors, complicating troubleshooting.

Risks and Mitigations

Data Security Vulnerabilities

Sensitive user information may be exposed if adequate security measures are not implemented.

Implement robust encryption methods for data at rest and in transit, conduct regular security audits, and follow industry best practices for data protection.

Integration Challenges

Difficulty in integrating with existing systems, such as Contentstack, could disrupt functionality.

Establish clear API documentation, conduct thorough integration testing with Contentstack, and allocate time for iterative adjustments based on feedback.

Compliance Risks

Failing to meet evolving data protection regulations could lead to legal issues and penalties.

Regularly review and update the authentication system to align with evolving data protection regulations, and maintain comprehensive compliance documentation.

Operational Downtime

System unavailability during maintenance or unexpected failures may disrupt user access.

Create a robust disaster recovery plan, perform regular backups, and conduct scheduled maintenance during low-usage hours to minimize user impact.

References:

- Supabase authentication docs: <https://supabase.com/docs>